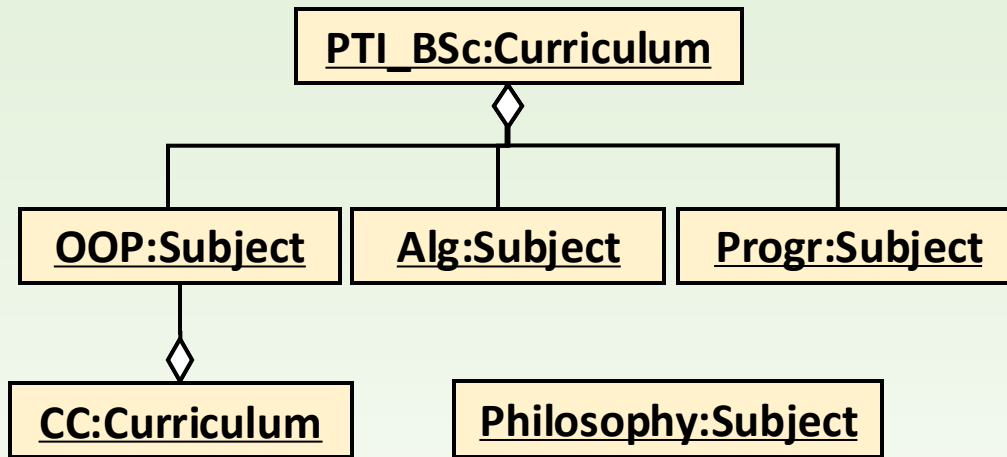
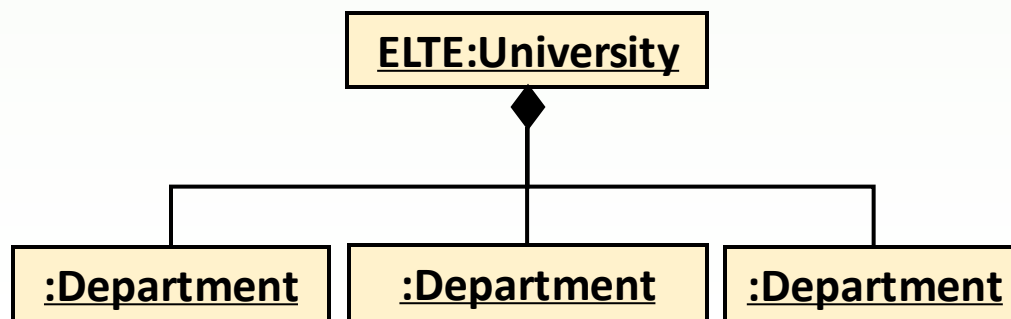


1/a A curriculum consists of subjects. A subject can be part of multiple curricula at the same time, and it does not have to be part of any curriculum.



1/b. A university has departments: there is no university without departments, and no department exists without a university. A department can only belong to one university and ceases to exist if its university is closed.



aggregation:

The same object can have multiple owners, and an owner can exist without an owned object

Curriculum

A subject can belong to 0, 1, ... curricula

Subject

Curriculum needs at least one subject

If multiplicity is not specified, it defaults to 1

composition: the container cannot exist without the contained, the contained can only be a part of one container

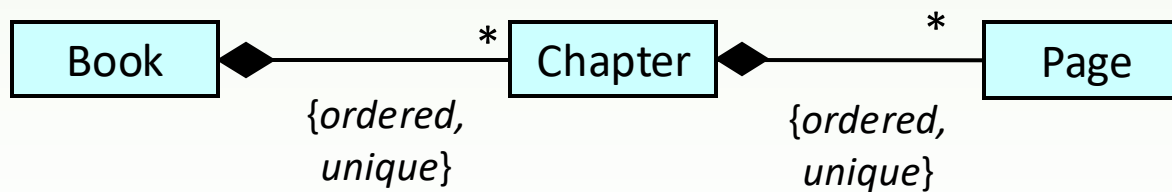
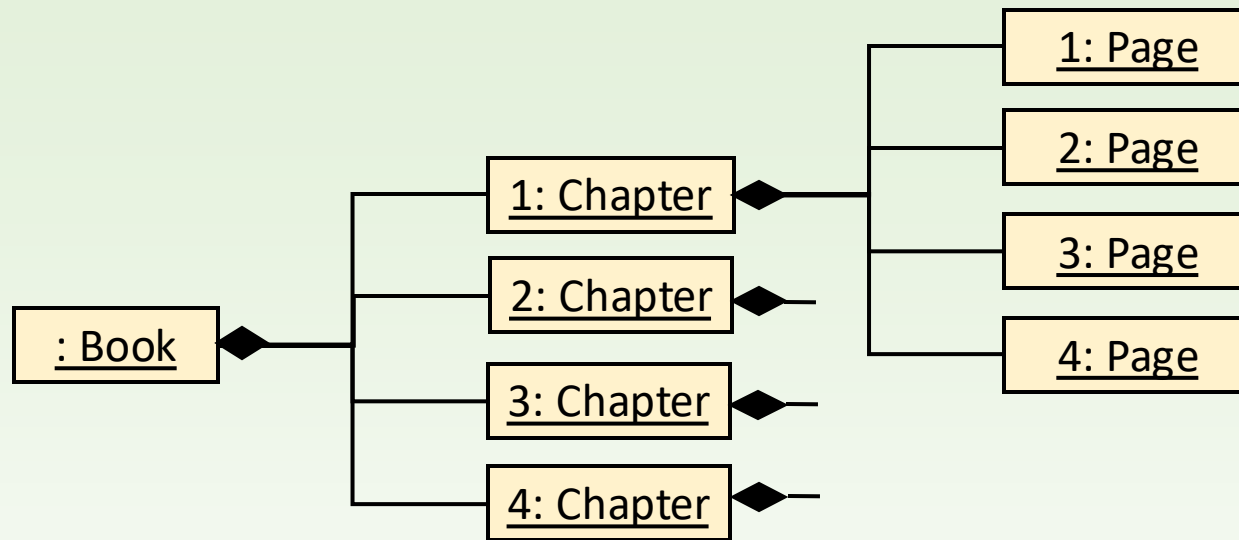
University

1 or 0..1 because of composition

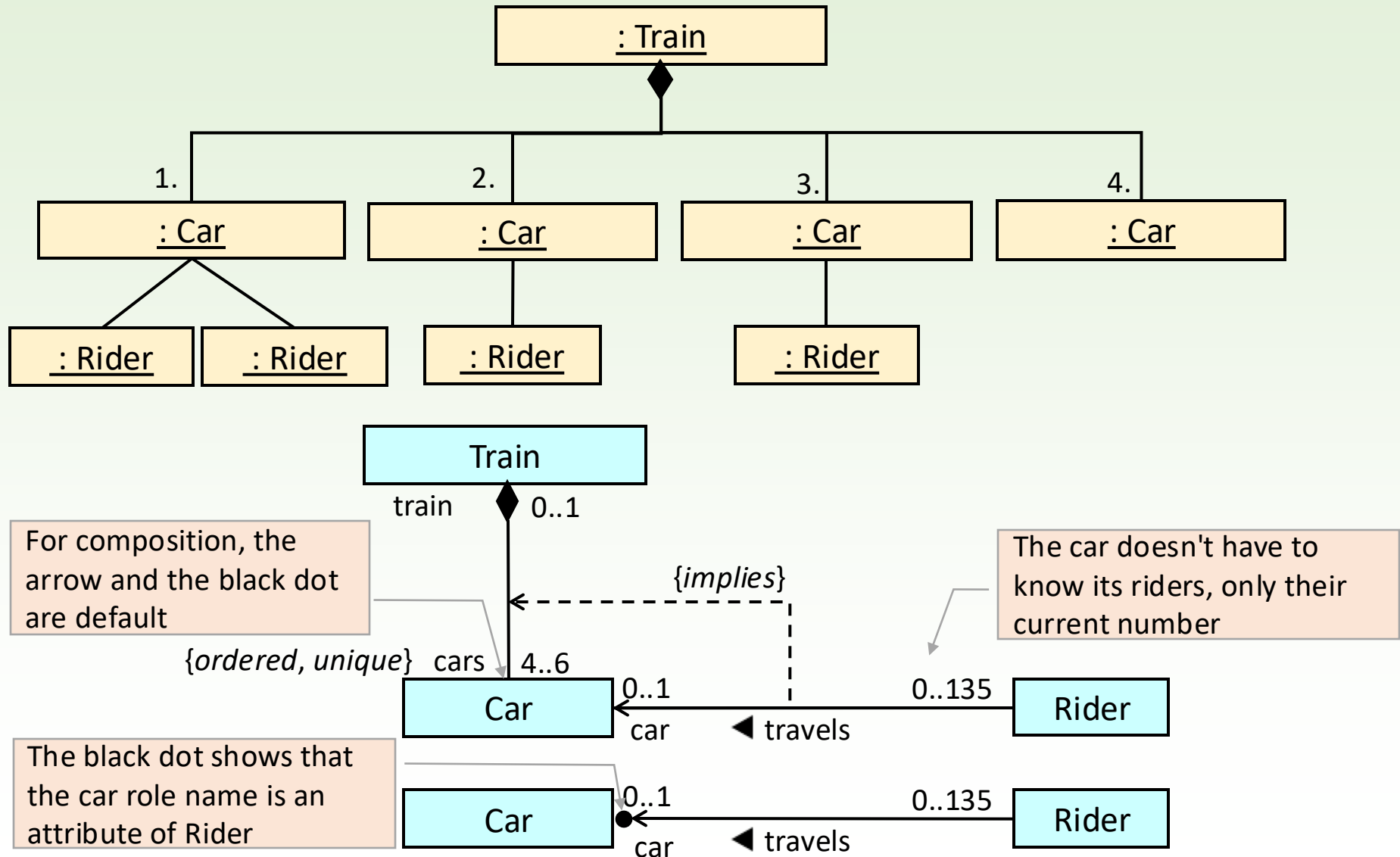
Department

This means 1..* because of the composition

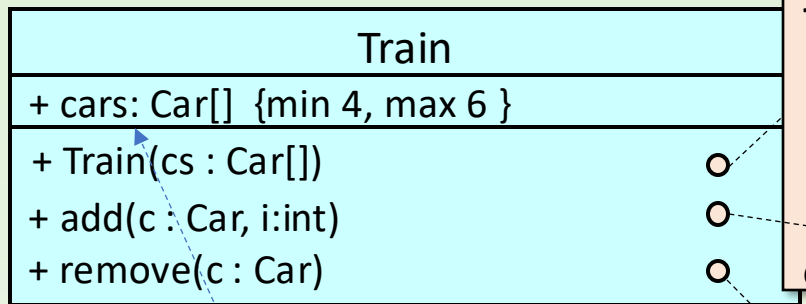
2. A book consists of at least one chapter, and a chapter contains at least one page.



3. A metro train consists of four, five, or six cars connected in a specific order. A passenger can board a metro car only if it is already part of a train. A metro car can accommodate up to 135 passengers.



Methods needed for relationships



```

if |cs|<4 or |cs|>6 then error endif
forall c in cs loop
  if c.train≠null then error endif
  if c in cars then error endif
  cars.add(c)
  c.train:=this
endloop

```

```

if |cars| = 6 then error endif
if c.train≠null then error endif
if c in cars then error endif
cars.insert(i, k)
c.train:=this

```

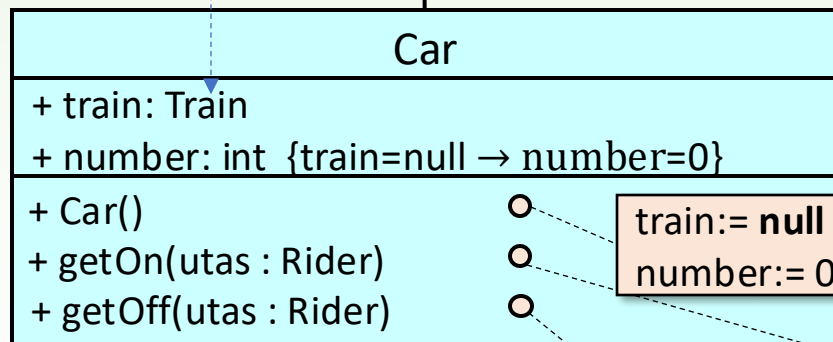
```

if |cars| = 4 then error endif
if c.number> 0 then error endif
cars.remove(c)
c.train:=null

```

train 0..1

cars 4..6{ordered, unique}



```

train:= null
number:= 0

```

```

if train= null then error endif
if number= 135 then error endif
if rider.car≠ null then error endif
number:= number+1
rider.car:=this

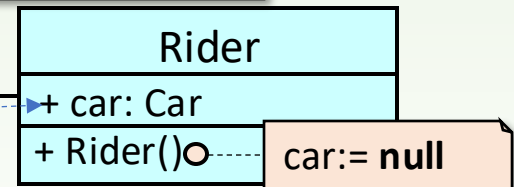
```

```

if rider.car≠ this then error endif
number:= number-1
rider.car:=null

```

0..1 travels 0..135
kocsi



```
car:= null
```

6. A company employs individuals in various positions. The job description includes the job start date, the job title, and the salary.

